

Shopify Sales Analysis Dashboard using Power BI

1. Business Objective

The objective of this project is to analyze Shopify sales data to understand:

- How much revenue the business earns
- Which products sell the most
- Who the valuable customers are
- How many customers return to purchase again
- How to increase sales and improve customer retention

The final goal is to build an interactive Power BI dashboard that helps business owners make better decisions using data.

2. Business Problem

The company has sales data, but it does not know:

- Which products generate the highest revenue
- Which customers buy frequently
- Whether customers return after their first purchase
- Which customer groups bring the highest value
- How sales change over time
- What actions can increase revenue and customer loyalty

The dashboard should answer these business questions.

3. Project Flow

Step 1 — Understand the Dataset

First understand:

- Number of tables
- Number of rows
- Number of columns
- Meaning of each column
- Primary Key

- Foreign Key

Examples of columns:

- Order ID
 - Customer ID
 - Order Date
 - Product
 - Quantity
 - Price
 - Sales Amount
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Step 2 — Data Cleaning

Check for:

- Missing values
- Duplicate rows
- Wrong data types
- Blank records
- Extra spaces
- Incorrect spellings
- Date format issues

Then clean the data.

Examples:

- Remove duplicates
 - Convert Date to Date datatype
 - Convert Quantity into Whole Number
 - Convert Price into Decimal
 - Remove unnecessary columns
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Step 3 — Data Modeling

Create relationships between tables.

If multiple tables exist:

Use a **Star Schema**

Example

Fact Table

- Orders

Dimension Tables

- Customer
- Product
- Date

Create a Calendar Table for time analysis.

Step 4 — Create DAX Measures

Create measures like:

- Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value
 - Average Quantity
 - Revenue Growth
 - Repeat Purchase Rate
 - Customer Lifetime Value
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Step 5 — Transaction Performance Analysis

Analyze

- Revenue by Month
- Revenue by Year
- Revenue by Product
- Revenue by Category
- Best Selling Products
- Worst Selling Products
- Peak Sales Time

- Sales Trend
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Step 6 — Customer Behaviour Analysis

Analyze

- New Customers
 - Returning Customers
 - Top Customers
 - Customer Spending
 - Customer Segments
 - RFM Analysis
 - Purchase Frequency
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Step 7 — Retention Analysis

Find

- Repeat Purchase Rate
 - Customer Lifetime Value
 - Average Days Between Purchases
 - Customer Retention
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Step 8 — Business Insights

Write insights like:

- Highest selling products
 - Lowest selling products
 - Best customer segment
 - Revenue trend
 - Customer retention trend
 - Highest CLV customers
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Step 9 — Business Recommendations

Suggest improvements based on your findings.

6. Deliverables

At the end of the project, you should submit:

1. Power BI Dashboard (.pbix)